Ruiyang Zhang

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Education

University of Macau — **Macau SAR**
Aug. 2024 - present
- Ph.D. in Faculty of Science and Technology
- Advisor: Professor Zhedong Zheng

Fudan University — **Shanghai, China**
Sep. 2017 - Jun. 2021
- B.S. in Computer Science and Technology

Research Interests

- Ultra-Long-Horizon Coding Agent, Agentic Post-Training, Efficient Reasoning

Selected Publications

For the full list, please refer to the Google Scholar Page.

Marathoner: Ultra-Long-Horizon Autonomous Intelligence
- Ruiyang Zhang, Jinpeng Ou, Yifan Xie, Jingang Zhou, Lirui Pan, Qingpei Guo, Zhedong Zheng
- In submission to **ICLR 2027**
- Build ultra-long-horizon coding agent through a comprehensive post-training pipeline, including Ultra-Long-Horizon Task Synthesis, Rejection Sampling Finetuning, and Reinforcement Learning. For data synthesis, we propose Multi-Task Chaining to produce even more challenging tasks. For RL, we introduce Later Stage Bonus Reward, which explicitly rewards valuable operation during later stage of execution, facilitating more effective ultra-long-horizon ability.

VSearcher: Long-Horizon Multimodal Search Agent via Reinforcement Learning
- Ruiyang Zhang, Qianguo Sun, Chao Song, Yiyan Qi, Zhedong Zheng
- **NeurIPS 2026**
- (1) Construct the RL infrastructure required to develop multi-turn tool-using multimodal search agent, including tool implementation, caching mechanism, and RL framework adaptation. (2) Train long-horizon, multi-turn multimodal search agent in real-world web environment via reinforcement learning. Propose a systematic post-training framework for multimodal search agents, including Iterative Injection-based Data Synthesis, Rejection Sampling Finetuning, and RL.

SketchThinker-R1: Towards Efficient Sketch-Style Reasoning in Large Multimodal Models
- Ruiyang Zhang*, Dongzhan Zhou*, Zhedong Zheng
- **ICLR 2026**
- (1) Devise Sketch-Mode Cold Start to instill initial sketch-style reasoning into base multimodal model. Train SketchJudge reward model to pose supervision on thinking style during the RL stage. Conduct Sketch-Thinking Reinforcement Learning to further generalize sketch-style reasoning. (2) Evaluation on benchmarks across various domains, showing that SketchThinker-R1 achieves over 64% reduction in thinking cost without compromising answer accuracy.

Uncertainty-o: One Model-agnostic Framework for Unveiling Epistemic Uncertainty in Large Multimodal Models
- Ruiyang Zhang, Hu Zhang, Hao Fei, Zhedong Zheng
- **EMNLP 2026**

VL-Uncertainty: Detecting Hallucination in Large Vision-Language Model via Uncertainty Estimation

- Ruiyang Zhang, Hu Zhang, Zhedong Zheng
- **Preprint**

Harnessing Uncertainty-aware Bounding Boxes for Unsupervised 3D Object Detection

- Ruiyang Zhang, Hu Zhang, Hang Yu, Zhedong Zheng
- **ICCV 2025**

Approaching Outside: Scaling Unsupervised 3D Object Detection from 2D Scene

- Ruiyang Zhang, Hu Zhang, Hang Yu, Zhedong Zheng
- **ECCV 2024**

Research Experiences

Ant Group — **Beijing, China**
June 2026 - present

- Research Intern, Ling Team, Foundation Model
- Mentor: Qingpei Guo
- Research Topic: Exploring and developing general pipeline for constructing ultra-long-horizon ability
- Project: Marathoner

IDEA — **Shenzhen, China**
Oct. 2025 - Mar. 2026

- Research Intern
- Mentor: Dr. Yiyan Qi
- Research Topic: Developing long-horizon multimodal search agent via reinforcement learning (RL)
- Project: VSearcher (NeurIPS 2026)

Shanghai AI Lab — **Shanghai, China**
May 2025 - Sep. 2025

- Research Intern
- Mentor: Dr. Dongzhan Zhou
- Research Topic: Leveraging RL to incentivize sketch-style reasoning in large multimodal models
- Project: SketchThinker-R1 (ICLR 2026)

Honors and Awards

- **Second Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM). 2019
- **Third Class Scholarship for Outstanding Students**, Fudan University. 2018 and 2019
- **First Prize**, Chinese Mathematical Olympiad in Jiangsu Province (CMO). 2016
- **First Prize**, National Olympiad in Informatics in Jiangsu Province (NOIP). 2014 and 2015